

# **Master Thesis — Progress Report**

## **Rewiring the market clock in Spanish electricity markets**

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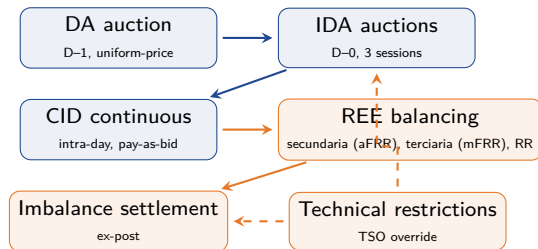
# Outline

- Motivation
- Market structure and the 2024–2025 reform sequence
- Economic mechanisms: theoretical anchors and our mechanism
- What we want to test, data, and bid-shape evidence
- Identification design and main result
- What is robust, what is not, and what is left

# Motivation

- Empirically assess the impact of Spain's 2024–2025 **market time unit (MTU15) reform sequence** on *firm strategic behaviour* and *market outcomes*.
- Why interesting: the reform changes the time-unit granularity of the wholesale market in three stages, with *transitional asymmetric-clock windows* where settlement and trading run on different clocks. A useful setting to study how granularity shapes firm conduct.
- Why it is hard: the reform sequence overlaps with confounders — gas-price normalisation (Dec 2023), the 2025-04-28 blackout and post-blackout *operación reforzada* (reinforced operation), the elimination of CNMC Rule 28.8 for bilateral contracts (Mar 2025).  
**Institutional awareness is essential.**
- This presentation: identification strategy, headline result on **dominant firms** (Iberdrola, Endesa, Naturgy, EDP) strategic intraday repositioning, and an honest account of what survives robustness and what does not.

# The Spanish wholesale market



- **OMIE (market operator)**: day-ahead, intraday auctions, continuous intraday.
- **REE (TSO)**: balancing markets (secundaria = aFRR, terciaria = mFRR, replacement reserve), imbalance settlement, technical restrictions.
- **Sequence**: forward markets get progressively closer to delivery; final adjustments via balancing.
- TSO can modify OMIE programmes when network or stability constraints bind.
- Firms have multiple instruments to take strategic positions across stages.

# The 2024–2025 reform sequence

- Traditionally MTU = **60 min** everywhere.
- **Jun 2024 — IDA reform:** 6 local Iberian sessions consolidated into 3 European IDA sessions (still 60-min granularity).
- Three MTU15 reform stages:
  - **Dec 2024 — ISP15:** imbalance settlement → 15 min.
  - **Mar 2025 — ID15:** intraday → 15 min.
  - **Oct 2025 — DA15:** day-ahead → 15 min.
- Two transitional *asymmetric-clock* windows: settlement runs at finer granularity than trading.
- Plus: **2025-04-28 blackout** → *operación reforzada*, continuous from Apr 28 onward.

| Period       | ISP       | ID        | DA        |
|--------------|-----------|-----------|-----------|
| Pre-Dec 2024 | 60        | 60        | 60        |
| Dec24–Mar25  | <b>15</b> | 60        | 60        |
| Mar25–Sep25  | 15        | <b>15</b> | 60        |
| Oct25 onward | 15        | 15        | <b>15</b> |

ISP15

*Dec 2024*

ID15

*Mar 2025*

DA15

*Oct 2025*

# Theoretical anchors and our mechanism

## Allaz & Vila (1993, JET) — forward markets dispel rents

Adding more forward markets prior to the spot stage commits Cournot firms to quantities and erodes withholding incentives. *Caveat:* A–V adds **more markets for the same good** (e.g., a second forward stage). Our reform **divides one good into four sub-goods** (an hour into 4 quarters). **Related intuition; not the same mechanism.**

## Ito & Reguant (2016, AER) — sequential markets and arbitrage

Firms with market power reposition across sequential markets (under-commit in DA, sell more in IDA). *Caveat:* I–R is about **sequencing across markets**; we are about **granularity within a market**. **Related, not equivalent.**

## Chang (2026, WP) — settlement vs dispatch period mismatch

When settlement period differs from dispatch period, averaging effects change bidding incentives. *Does not apply to DA15/ID15:* settlement and dispatch are both 15-min — no averaging. Only the ISP15-with-DA60 transition window had averaging effects, and only on *quantities* (imbalance settlement averaged within the hour).

## Our preliminary mechanism — within-market granularity (what we test)

Splitting an hour into 4 quarters that clear independently lets firms bid different prices and quantities across quarters of the same hour. **Especially valuable when residual demand varies steeply within the hour** (morning ramps, evening peaks). In flat hours where within-hour demand is constant, granularity has no economic content. **To be formalized!**

# What we want to test, and how

**Our mechanism predicts.** Dominant firms exploit within-hour granularity in *critical* hours (steep within-hour ramps in residual demand) but not in *flat* hours (constant within-hour profile). The reform should differentially affect critical vs flat hours.

## Outcomes:

- $q_2$  = dominant firms' *voluntary* IDA repositioning per firm-day-hour (signed PIBCIE), MWh. **Headline outcome.**
- $q_1$  = dominant firms' day-ahead auction-cleared sell, MWh per firm-hour.
- Bid-shape: tranche count, ladder slope, qty-weighted price (post-MTU15-DA only).

**How we define critical and flat hours.** For each hour-of-day  $h$ , compute  $\sigma_{\text{within}}^2(h)$  = **the variance of conventional production (load – wind – solar) across the four 15-min quarters of hour  $h$** , averaged over days. High  $\sigma_{\text{within}}^2$  means dispatchable units need to ramp a lot *within* the hour — exactly where within-hour granularity has economic content.

- **Treated** = critical hours  $h \in \{7, 8, 16, 17, 18\}$ . Top-5 by  $\sigma_{\text{within}}^2$  (morning ramp + evening peak).
- **Control** = flat hours  $h \in \{3, 4, 5\}$ . Bottom of  $\sigma_{\text{within}}^2$  (pre-dawn).

**Why flat hours are a valid counterfactual.** If the granularity reform has *zero economic content* in flat hours (because there is nothing within the hour to slice strategically), then comparing critical-hour behaviour to flat-hour behaviour identifies the mechanism. *This is a hypothesis I need to defend with a small theoretical model showing that, in equilibrium, the reform has no effect for flat hours — ongoing.*

# Data

**Coverage:** 8 years (2018-01 → 2026-04), at native 60-min/15-min granularity throughout.

## Three sources:

- **OMIE (Iberian market operator):** prices and *bid-level* data for DA, IDA, continuous; per-firm and per-unit programmes. ~5.7 billion rows (~413 GB raw, ~198 GB processed).
- **ENTSO-E Transparency Platform:** imbalance volumes and prices, balancing settlement, generation by type/unit, load and renewable forecasts. All system-level fundamentals come from here.
- **ESIOS (REE):** Spanish per-segment settlement, real-time prices, balancing offers. **API token requested 3 weeks ago; still pending due to technical issues on the operator's side.** Reserves attribution at firm level partially blocked.

## Engineering principles:

- Pipelines are **idempotent**: re-running any step produces the same output and includes new data — supports replication.
- 25+ data families, each with its own parser and tests.
- Empirical findings tracked formally (alive / wounded / dead) with a documented audit trail.

**Honest caveat.** Most of the empirical findings in the audit log are descriptive rather than carrying clear economic interest — e.g., the structural break in Q1 2024 nuclear bilateral share. The presentation focuses on the small subset that has both economic content and clean identification.

## Bid-shape: dominant firms exploit granularity (justifies the design)

Direct evidence at the bid level that operators *actually use* within-hour granularity post-MTU15-DA — this justifies using flat hours as a valid counterfactual.

**What we observe.** For dominant-firm CCGT in critical hours (post-MTU15-DA):

- More price tranches per quarter (+8% vs flat hours).
- Steeper ladder slope (+13%).
- Lower qty-weighted average price.
- P90 of **CV** (coefficient of variation = std dev / mean) of average price across the 4 quarters of an hour: **7.9% in critical vs 1.6% in flat (4.9× wider).**

**Interpretation.** Operators bid a richer ladder in critical hours: low-price inframarginal tranches at the bottom (push the qty-weighted average down, lock clearing) plus a steeper high-price tail (capture scarcity rent). They are unambiguously exploiting within-hour granularity.

**Tech selectivity.** Concentrated in CCGT and (mildly) hydro. Wind, solar, nuclear show no engagement. Crucially: many plants **mechanically repeat their previous hourly bid identically across all 4 quarters** — not exploiting the granularity at all.

**Why this matters for the design.** Bids confirm operators *can* and *do* use within-hour granularity in the right hours. Where granularity is exploited (critical), we should see effects on quantities; where not (flat), we should not.

# The DiD specification

**Panel.** The unit of observation is (firm  $f$ , day  $d$ , hour  $h$ ). We aggregate across the 4 quarters within an hour. So the outcome is  $q_{fdh}$ .

**Specification (very preliminary):**

$$q_{fdh} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 (\text{crit}_h \times \text{post}_d) + \gamma_f + \delta_{\text{DOW}(d)} + \varepsilon_{fdh}$$

$\beta_3$  is the DiD coefficient.  $\gamma_f$  are firm fixed effects (Iberdrola, Endesa, Naturgy, EDP);  $\delta_{\text{DOW}}$  are day-of-week fixed effects. Cluster-robust SE by date  $d$ .

**Window in each regression:**

- **Pre** = pre-IDA: 2018-01-01  $\rightarrow$  2024-06-13.
- **Post** = DA15/ID15: 2025-10-01  $\rightarrow$  present.
- Intermediate regimes (3-sess, ISP15-win, DA60/ID15) handled separately when needed.

**Why the DiD removes time-varying confounders:**

- Critical and flat hours come from the *same dates*  $\Rightarrow$  daily shocks (weather, demand level, fuel prices, regulatory announcements, macro conditions) absorb mechanically in the within-day comparison.
- Pre-IDA crit-flat differential serves as the parallel-trends sanity check.
- The contrast we identify is *mechanism-targeted*: hours where granularity should bite vs hours where it should not.

**Heterogeneity.** Most of the response carried by Naturgy (CCGT + Hydro). Iberdrola contributes through nuclear (caveat: allocation artefact). Endesa and EDP near-zero. Different technologies respond very differently — consistent with the bid-shape evidence on the previous slide.

## Headline result: dominant firms reposition more in critical hours

Same panel and spec as previous slide: outcome  $q_{fdh}$  (signed IDA repositioning, MWh per firm-day-hour);  $\text{crit}_h = 1$  for  $h \in \{7, 8, 16, 17, 18\}$ , 0 for  $h \in \{3, 4, 5\}$ ;  $\text{post}_d = 1$  from 2025-10, 0 for pre-IDA.

|                                                         | Estimate | SE    | p-value               |
|---------------------------------------------------------|----------|-------|-----------------------|
| $\beta_1$ ( $\text{crit}_h$ , pre-IDA crit–flat gap)    | +41.21   | 3.36  | $1.6 \times 10^{-34}$ |
| $\beta_3$ ( $\text{crit}_h \times \text{post}_d$ , DiD) | +58.67   | 12.18 | $1.5 \times 10^{-6}$  |
| $\beta_1 + \beta_3$ (DA15/ID15 crit–flat gap)           | +99.80   | —     | —                     |

$N = 78,177$  firm-day-hours; cluster-robust SE by date.

**Sample-size caveat.**  $N \approx 80k$  is huge —  $p$ -values are mechanically tiny even for modest effects. Significance thresholds should be tightened accordingly (e.g.,  $p < 10^{-4}$  instead of 0.05).

**Why these hours.** Adding more hours dilutes the result: midday hours (h10–h15) have the *opposite-sign* response under DA15/ID15 (firms reposit *less* there, due to solar abundance). Top-5 critical vs bottom-3 flat isolates the rising-production cluster cleanly.

**Interpretation.** The outcome  $q_2$  (signed IDA repositioning above DA commitment) is the empirical signature **Ito & Reguant (2016) use to detect strategic intra-day arbitrage**: positive  $q_2$  means firms under-commit in DA and resell in IDA. We adopt their measure but ask a *different* question: **does this strategic IDA repositioning grow more in critical hours after the reform splits each hour into 4 sub-periods?** The DiD coefficient  $\beta_3 = +59$  says yes. Mechanism: **the reform lets firms bid different prices/quantities for each quarter of an hour. This pays off where the within-hour production profile is steep — so firms exploit the ramps. Flat hours have no within-hour variation, so no effect.**

## Robustness checks (in words)

- **Same-calendar-month restriction.** Restricting pre-IDA to the same months as the post window (Oct–May) leaves the result at 96% of the magnitude. Calendar mix not the driver.
- **Baseline-window sensitivity.** Across 5 narrower pre-IDA windows, the within-DA15/ID15 fact (+100) is constant; reform-attribution component ranges +27 to +133. **To think about:** pre-trend in the crit-flat gap is not stable.
- **Per-hour decomposition.** Treating each hour individually shows two clusters with opposite signs (ramps + peak positive; midday negative). Justifies the 5-hour critical set.
- **Conditional parallel trends.** Wind + solar + demand  $\times$  hour controls absorb only  $\sim 13\%$  of the pre-trend. *Does NOT hold.* **To think about:** an unobserved time trend (CCGT retirement, regulatory anticipation) is at play.
- **Bid-shape and extensive-margin.** Unit-level confirmation: richer bid ladders in critical hours (CCGT only); 13 CCGT units inactive in the asymmetric-clock window, return under *operación reforzada*.

# What is robust, what is not

## Robust:

- The within-DA15/ID15 fact: +100 MWh more IDA repositioning in critical vs flat hours per firm-hour.
- Bid-shape rich-ladder pattern in CCGT post-MTU15-DA.
- Two-cluster mechanism: ramp + peak positive; midday negative.

## Not robust:

- The reform-attributable component (DiD  $\delta$ ) is a range, not a point estimate. Pre-trend instability is real.
- Conditional parallel trends with renewable + demand controls does not hold.
- Heterogeneity: Naturgy carries most of the result; cross-firm response is uneven.

**The theoretical-model gap.** The design needs a stylised IO model showing that, in equilibrium, the within-market granularity reform is a no-op for flat hours — this is what justifies using them as control. The model is anchored in *our* mechanism (bidding differently across the 4 quarters of an hour, exploiting within-hour ramps), not directly in Allaz–Vila or Ito–Reguant. **Out of scope:** a fully estimated structural auction model.

# What is left

- **Stylised theoretical model** that justifies flat hours as control — the foundation of the empirical design.
- **Better identification strategy.** The clean flat-hour design came together yesterday; there is more thinking to do.
- Think about the demand side more — the current mechanism and design are supply-side focused.
- Include the continuous market in the analysis.
- Additional robustness checks once the mechanism interpretation is locked.
- Per-segment settlement once the ESIOS API token clears, but not necessary a bottleneck for the main result.

**Thank you!**